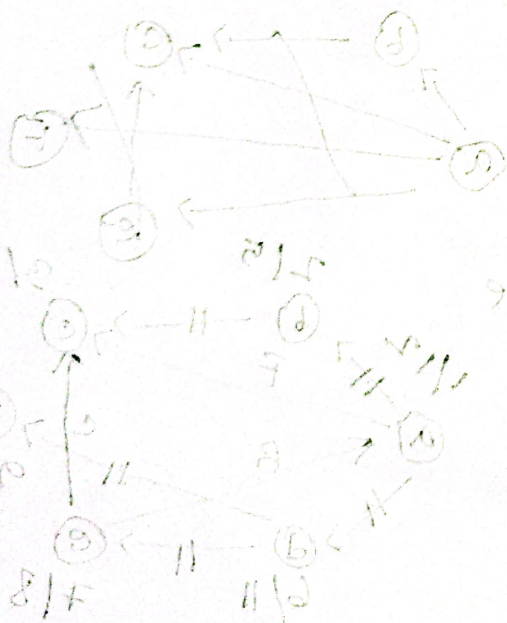


# Classification of edge in DFS

1. Tree Edge  $\rightarrow$  DFS apply করার পর যে DFS করতে পারবে
2. Forward Edge  $\rightarrow E(u, v)$  যেখানে starting time  $u < v$   
এবং  $u$  to  $v$  path থাকবে  
 \*  $u$  থেকে যেখানে থেকে তার বের হবে  
 \*  $v$   $u$  যেখানে তার ফিরবে  
 $E = u$   
 $F = u$
3. Back Edge  $\rightarrow E(u, v)$  যেখানে starting time  $u > v$   
এবং  $v$  to  $u$  path থাকবে  
 $E = v$   
 $F = u$
4. Cross Edge  $\rightarrow u > v$  but  $v$  to  $u$  path না  
দেখানো নাই  
 $E = v$   
 $F = u$



1 to 2 is

$F = 1$

$E = 1$

$u > v$

cross edge

1 to 3 is

$F = 1$

$E = 1$

1 to 4 is

cross edge

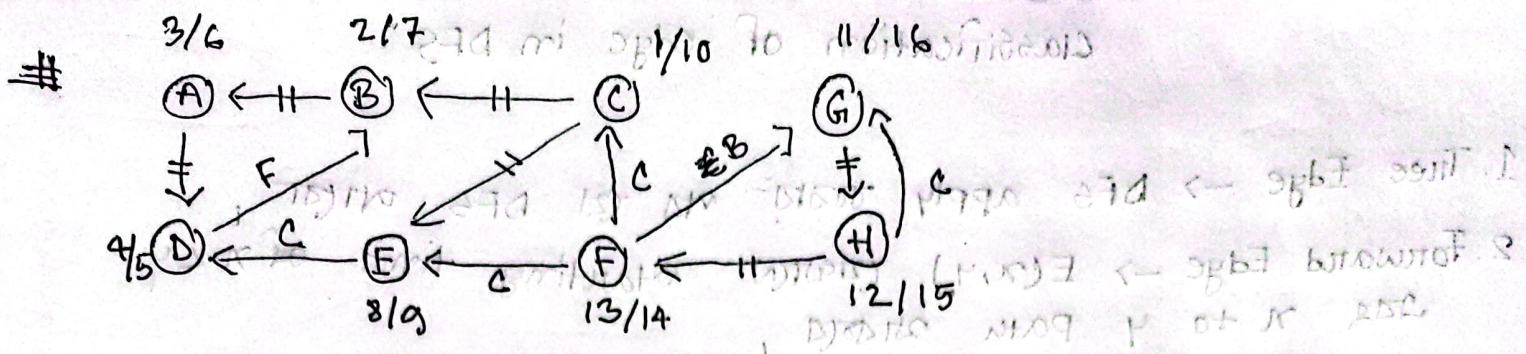
1 to 5 is

$F = 1$

$E = 1$

1 to 6 is

cross edge



H → Here means tree edge.

\* D to B  
 $x = 4$   
 $y = 2$   
 $\therefore x > y$  and B to D  
 Path available  
 Forward edge  
 Back

\* E to D  
 $x = 8$   
 $y = 4$   
 $\therefore x > y$  and  
 D to E has  
 no path  
 $\therefore$  cross edge

\* F to G  
 $x = 13$   
 $y = 1$   
 $\therefore x > y$  and No  
 path  
 $\therefore$  cross edge

\* F to E  
 $F = 13$   
 $E = 8$   
 $x > y$  and ~~F~~ No path  
 $\therefore$  cross edge

\* F to G  
 $F = 13$   
 $G = 11$   
 $\therefore x > y$  No path  
 $\therefore$  cross edge

\* H to G  
 $H = 12$   
 $G = 11$   
 $x > y$  No path  
 cross edge  
 Back

\* E to C  
 $E = 7$   
 $C = 3$   
 $\therefore x > y$  and No path  
 $\therefore$  cross

\* A to C  
 $A = 1$   
 $C = 3$   
 $x < y$   
 $\therefore$  Forward edge

\* E to A  
 $E = 7$   
 $A = 1$   
 $\therefore x > y$  and has path  
 Backward edge

